Human Input Limits: Technical Worksheet

Grade 8 · Worksheet · About 25 minutes
Work with pen and paper. Use color pencils if you like. You do not need a computer or the internet.

Name: _____

Grade/Section: _____

Date: _____

How to use this worksheet

- Do the parts **in order** from top to bottom.
- Each part has a **time** so you know how long to spend.
- Read every instruction before you write. You can do this on your own.

Part 1 – What you will learn (about 3 minutes)

What you will learn: Human input limits are boundaries of attention, memory, motor precision, and speed. Interfaces that ignore limits cause errors, fatigue, and abandoned tasks.

Do this:

1. **Describe a time** you made an input error because you were tired or rushed.

   _____

2. **Circle** a human limit: Working memory size · Infinite perfect attention

Part 2 – Attention and working memory (about 5 minutes)

People hold only a few chunks in **working memory**. Long codes, multi-field forms without save, and hidden instructions overload users.

> **Real-life scenario:** A password reset shows a 16-character code for 10 seconds with no copy button. Users mis-type it repeatedly.

Discuss with a partner or reflect in writing: How would you redesign that step to respect memory limits?

Do this: Circle **Yes** or **No**.

Design	Respects limits?
Show instructions next to the field	Yes · No
Force memorizing 5 rules on a prior page only	Yes · No
Allow paste for one-time codes	Yes · No

Part 3 – Motor precision and Fitts-like ideas (about 5 minutes)

Small targets far from the cursor are slower and more error-prone. Touch targets need size and spacing; keyboard focus order matters too.

> **Real-life scenario:** A “Delete account” button sits tiny next to “Save.” Tired users hit Delete.

Critical thinking: What spacing/label changes reduce that risk?

Do this:

1. **List two UI choices** that reduce motor errors.

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2. **Circle** safer placement: Destructive action separated + confirmed · Tiny delete beside save

Part 4 – Fatigue, speed, and error rates (about 5 minutes)

As fatigue rises, speed may stay high while **error rate** climbs. Good systems allow breaks, autosave, and undo rather than punishing humans for being human.

> **Real-life scenario:** During a 90-minute data entry drill with no breaks, late entries show swapped digits.

Discuss with a partner or reflect in writing: Which system feature would best protect data quality here?

Do this: Circle **Yes** or **No**.

Feature	Helps under fatigue?
Autosave every minute	Yes · No
No undo allowed ever	Yes · No
Progress saved if the tab closes	Yes · No

Part 5 – Limit-aware redesign (about 5 minutes)

Do this: Pick a painful school login/form. Sketch a redesign that respects one attention limit and one motor limit.

State the limit you designed for in one sentence.

Part 6 – Check what you learned (about 2 minutes)

Checklist. Put a ✓ in the box when it is true for you.

☐ I can name key **human input limits**.

☐ I relate working memory to form design.

☐ I consider target size/placement for motor limits.

☐ I connect fatigue to error rates.

☐ I can sketch a limit-aware redesign.

When all boxes are checked, you have reviewed the main ideas from this worksheet.

Worksheet on Human Input Limits (Technical). Central topic: Human input limits. Main technology area: Human factors and HCI.